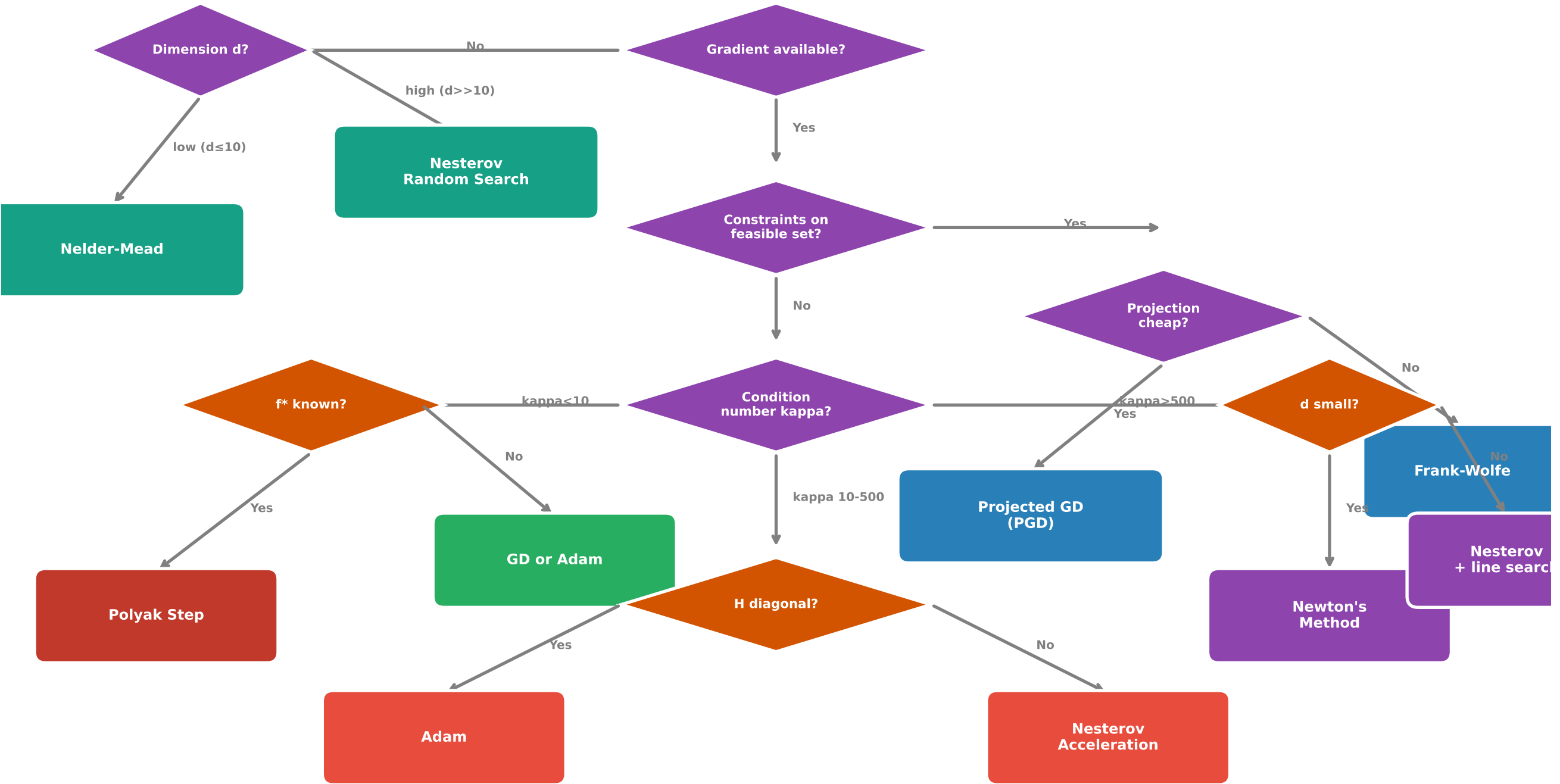


Algorithm Selection Decision Tree for Optimisation Problems



Convergence Rates

- Large-scale: Mini-batch SGD (with Adam or Nesterov momentum)
- Large-batch stability with adaptive scaling for best results
- Polyak: $O(1/k)$, no L, μ needed
- GD: $O(\rho^k)$, $\rho = (\kappa - 1) / (\kappa + 1)$
- Adam/Nesterov: $O(1/k^2)$ or $O(\rho^{k/2})$
- Newton: $O(\rho^{2^k})$ quadratic
- PGD/FW: $O(\rho^k)$ or $O(1/k)$
- Zeroth-order: $O(d^2/k)$ or empirical